

LAMP

ASP Rule Induction and Optimization with LLMs

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LAMP LLM to ASP Modeling Protocol

Task

ASP rule induction. Given facts F and a target answer set A^* , find a small program Π with

$$AS(F \cup \Pi) = \{A^*\}.$$

Bounded unary normal rules. Humans find this hard.

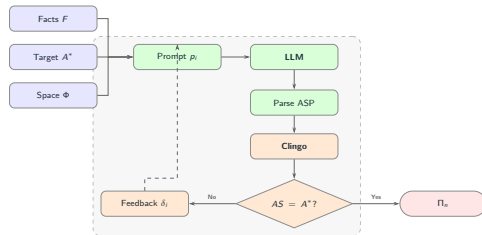
Loop

- LLM proposes rules from F , A^* , schema bounds.
- CLINGO verifies the stable model.
- Mismatches feed back. Up to 3 attempts.
- No fine tuning. Solver decides correctness.

Setup

120 synthetic benchmarks varying predicates, terms, facts, body literals, negation, derived atoms. ILASP baseline with 420 s timeout.

Iterative Loop



Results across 120 benchmarks

System	Final acc.	Failures
<i>gpt-5-mini</i>	99.2%	1
<i>gpt-oss:20b</i>	98.3%	2
<i>qwen3-coder:30b</i>	64.2%	43
ILASP	97.5%	3